Tutorial 3

Exercise 1:

Consider the `Vehicle` table below from a logistics management database tracking fleet distribution:

vehicle_id	model_name	capacity_tons	primary_driver	last_maintenance
TRK-201	Volvo FH16	25	Anderson, T.	2026-03-15
VAN-045	Ford Transit	2	Martinez, L.	2026-05-22
TRK-108	Scania R500	18	Anderson, T.	2026-01-10
VAN-012	Ram ProMaster	3	Nguyen, K.	2026-06-02

Questions:

- **a.** What is the degree of this table?
- **b.** What is the possible domain for the following fields?
  - `vehicle_id`
  - `primary_driver`
  - `capacity_tons`

Exercise 2:

Correct the design flaws in the following initial table schemas to improve data integrity:

- **a. Vehicle Table:** `Vehicle (vehicle_id, model_name, capacity_tons, primary_driver, last_maintenance)`

- **b. Medical_Clinics:** `Clinics (Clinic_ID, Clinic_Name, Location_City_State, Lead_Physician_Full_Name)`
- **c. Corporate_Assets:** `Assets (Asset_ID, Asset_Name, Description, Unit_Value_USD_EUR)`
- **d. Academic_Research:** `Research_Paper (DOI_Number, Paper_Title, Co_Authors_Names, Publication_Year)`

Exercise 3:

Provide a clear real-world database attribute example domain for each of the following data types:

- **String of characters:**
- **Integer:**
- **Decimal number:**
- **Date:**
- **Boolean:**

Exercise 4:

Assume a scenario where a database needs to manage the sales, pre-orders, and inventory of an independent comic book store.

Identify:

- **a.** The main entities
- **b.** The relationships between these entities